

Explaintool HHGR Evaluation Report

dataset: contract_act_gold

document_id: 0940d367554383c5

top_k: 5

items_evaluated: 10

seed: 42

elapsed_seconds: 0.229

generated_at: 2026-08-05T17:04:02Z

Retrieval Accuracy & Latency by System

System	items	Recall@K	Precision@K	Hit Rate@K	MRR	MAP	NDCG@K	Mean Latency (ms)	p50 Latency (ms)	p95 Latency (ms)	Throughput (ops/sec)
Baseline	10	0.8000	0.1800	0.8000	0.7500	0.7500	0.7631	2.5130	2.2520	4.2550	397.9800
System A	10	0.8000	0.1800	0.8000	0.7000	0.6750	0.7139	0.3800	0.3720	0.4590	2629.5000
System B	10	0.9000	0.2000	0.9000	0.7583	0.7283	0.7781	0.0260	0.0230	0.0500	38610.0400
System C	10	0.8000	0.1800	0.8000	0.6500	0.6200	0.6743	0.1900	0.1850	0.2990	5268.7000
System D	10	0.8000	0.1800	0.8000	0.7000	0.6750	0.7139	0.4110	0.4000	0.4950	2433.0900

HHGR Explainability Metrics (per query)

item_id	Citation Accuracy	Hierarchy Correctness	Graph Path Accuracy	Provenance Completeness	Evidence Coverage	Counter-Authority F1	confidence
ica-001	1.0000	1.0000	1.0000	0.9500	0.9444	1.0000	0.8301
ica-002	1.0000	1.0000	1.0000	0.9500	1.0000	1.0000	0.7094
ica-003	1.0000	1.0000	1.0000	0.9643	0.6667	1.0000	0.7794
ica-004	1.0000	1.0000	1.0000	0.9643	0.5625	1.0000	0.7183
ica-005	1.0000	1.0000	1.0000	0.9643	0.7143	1.0000	0.7217
ica-006	1.0000	1.0000	1.0000	0.4357	0.0000	1.0000	0.4281
ica-007	0.5000	1.0000	1.0000	0.9643	0.8000	1.0000	0.7285
ica-008	1.0000	1.0000	1.0000	0.4500	0.5000	1.0000	0.6331
ica-009	1.0000	1.0000	1.0000	0.9500	1.0000	1.0000	0.7457
ica-010	1.0000	1.0000	1.0000	0.9500	0.6364	1.0000	0.4997

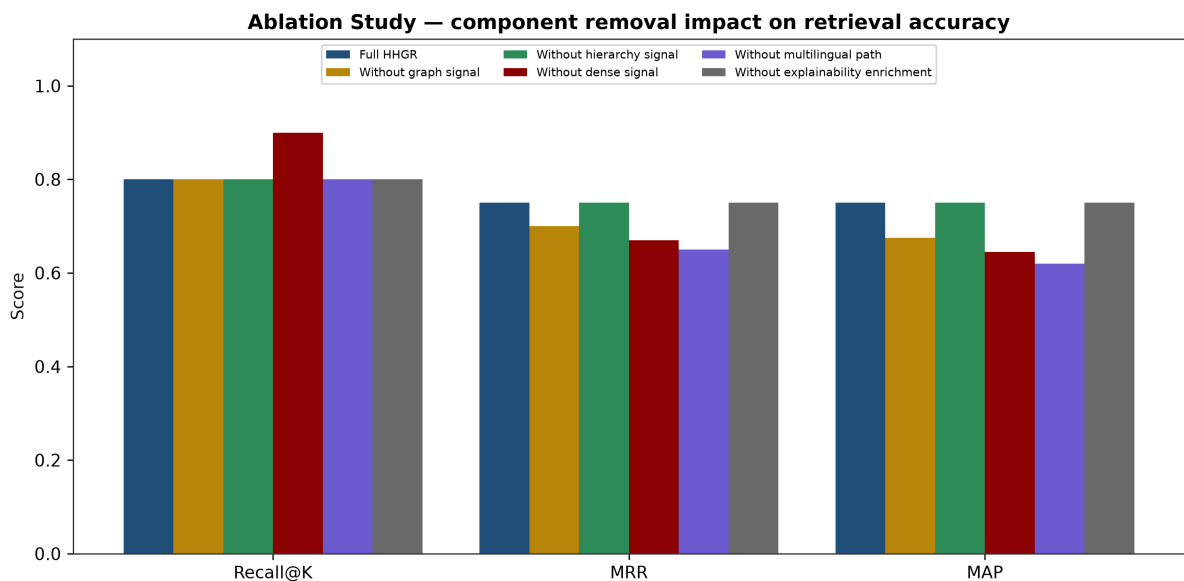
RAGAS-style Generation Metrics (offline surrogate)

system	item_id	Faithfulness	Answer Relevancy	Context Recall	Context Precision	Answer Correctness
hhgr	ica-001	1.0000	0.8081	1.0000	0.5731	0.5520
hhgr	ica-002	1.0000	0.7542	1.0000	0.4711	0.2549
hhgr	ica-003	1.0000	0.7864	1.0000	0.4804	0.4495
hhgr	ica-004	1.0000	0.7842	1.0000	0.4517	0.5843
hhgr	ica-005	1.0000	0.8563	1.0000	0.5079	0.3928

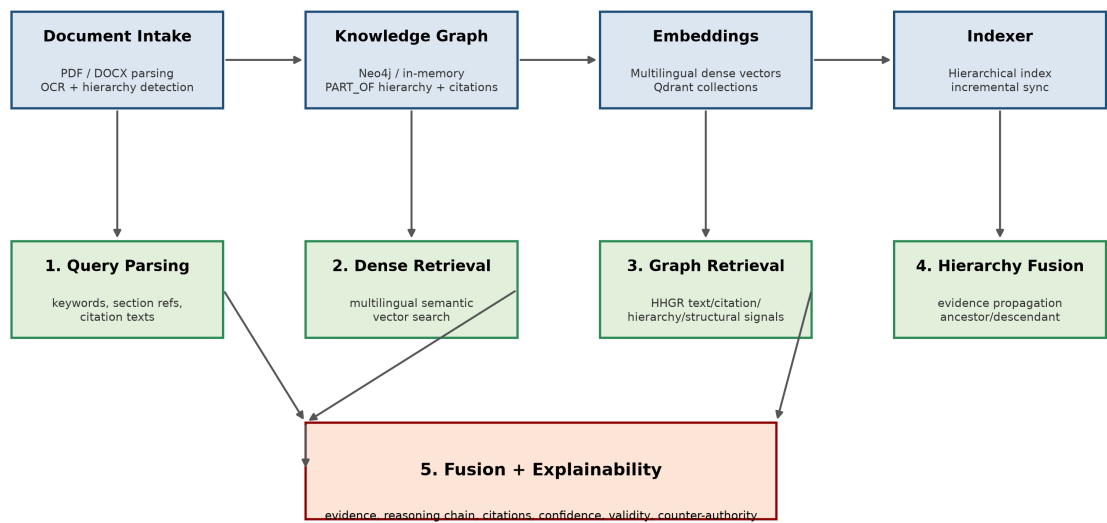
system	item_id	Faithfulness	Answer Relevancy	Context Recall	Context Precision	Answer Correctness
hhgr	ica-006	1.0000	0.6211	0.0000	0.0510	0.2331
hhgr	ica-007	1.0000	0.7548	0.5000	0.5376	0.4694
hhgr	ica-008	1.0000	0.8524	0.0000	0.4771	0.6262
hhgr	ica-009	1.0000	0.7178	1.0000	0.2542	0.4984
hhgr	ica-010	1.0000	0.8926	1.0000	0.3881	0.3366
naive_rag	ica-001	1.0000	0.8484	1.0000	0.5731	0.5714
naive_rag	ica-002	1.0000	0.8000	1.0000	0.4748	0.2748
naive_rag	ica-003	1.0000	0.8329	1.0000	0.4663	0.4634
naive_rag	ica-004	1.0000	0.8000	1.0000	0.4359	0.6186
naive_rag	ica-005	1.0000	0.8981	1.0000	0.5037	0.3976
naive_rag	ica-006	1.0000	0.6708	1.0000	0.0510	0.2518
naive_rag	ica-007	1.0000	0.8030	1.0000	0.5329	0.4534
naive_rag	ica-008	1.0000	0.8932	1.0000	0.4746	0.6672
naive_rag	ica-009	1.0000	0.7276	1.0000	0.2562	0.5192
naive_rag	ica-010	1.0000	0.9125	1.0000	0.3881	0.3455

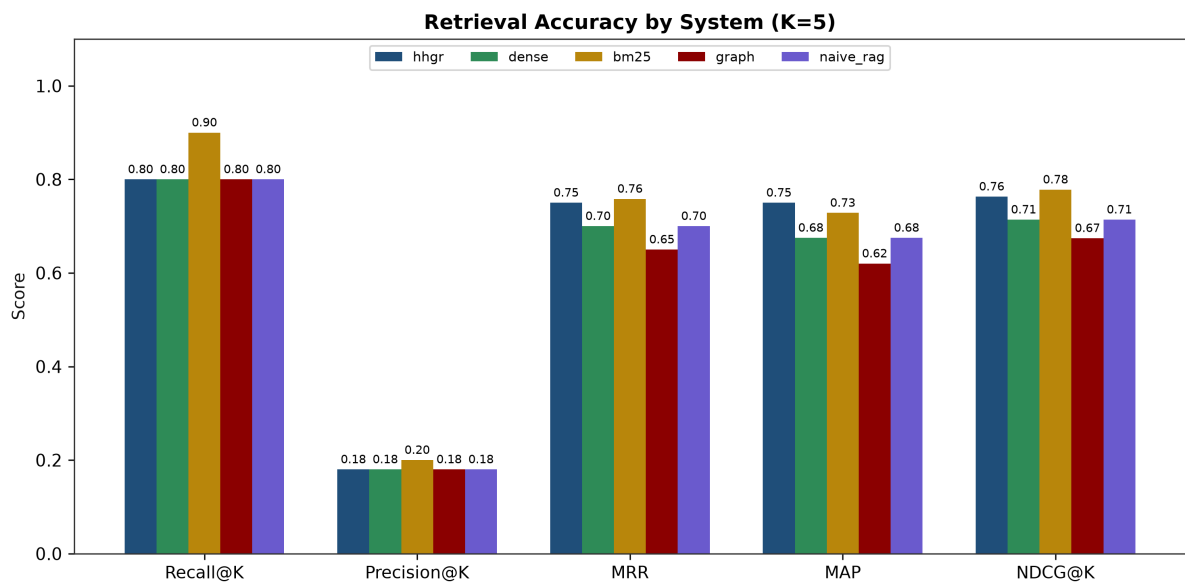
Ablation Study

ablation	label	Recall@K	MRR	MAP	NDCG@K	explain_citation_accuracy
full	Full HHGR	0.8000	0.7500	0.7500	0.7631	0.9500
no_graph	Without graph signal	0.8000	0.7000	0.6750	0.7139	0.9500
no_hierarchy	Without hierarchy signal	0.8000	0.7500	0.7500	0.7631	0.9500
no_dense	Without dense signal	0.9000	0.6700	0.6450	0.7157	0.9500
no_multilingual	Without multilingual path	0.8000	0.6500	0.6200	0.6743	0.7500
no_explainability	Without explainability enrichment	0.8000	0.7500	0.7500	0.7631	None

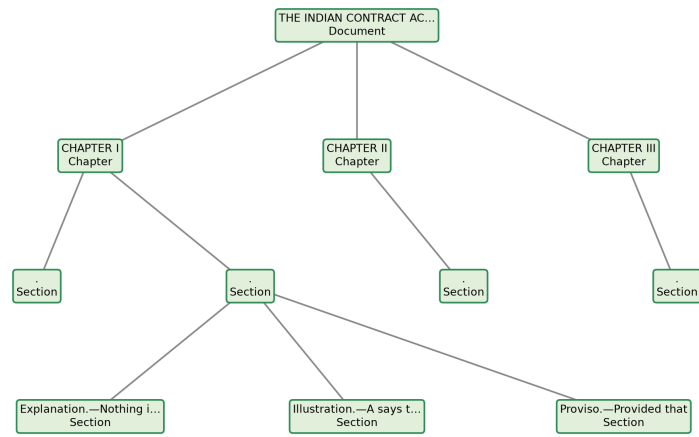


Explaintool: Explainable Multilingual Hierarchical Graph-RAG (HHGR) Architecture

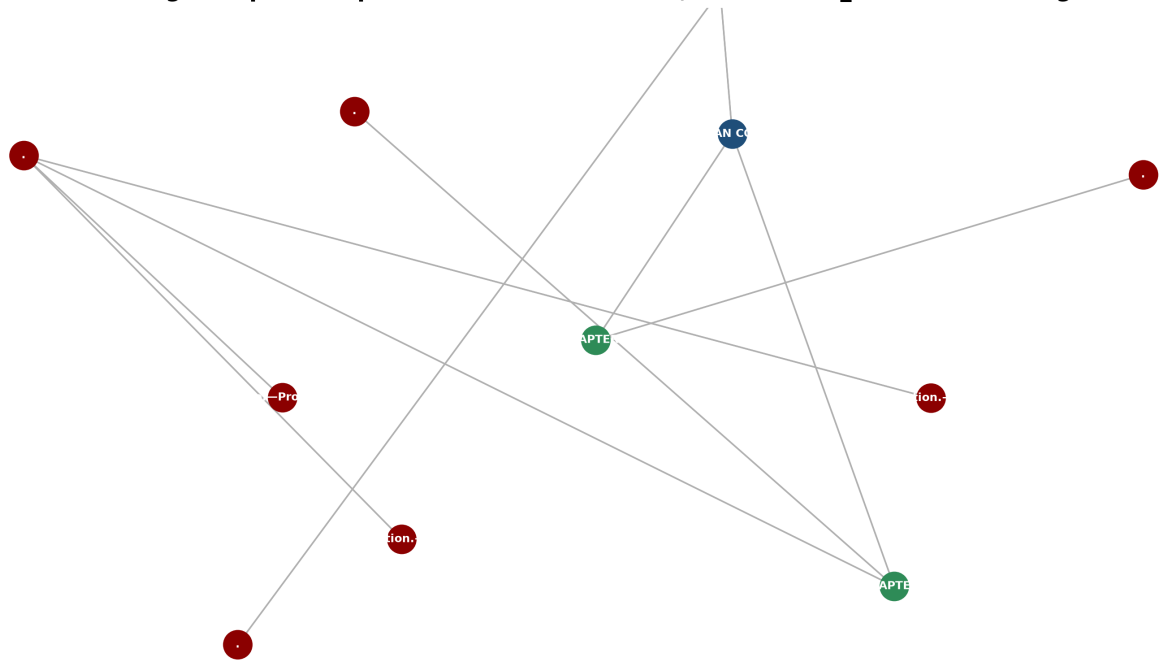


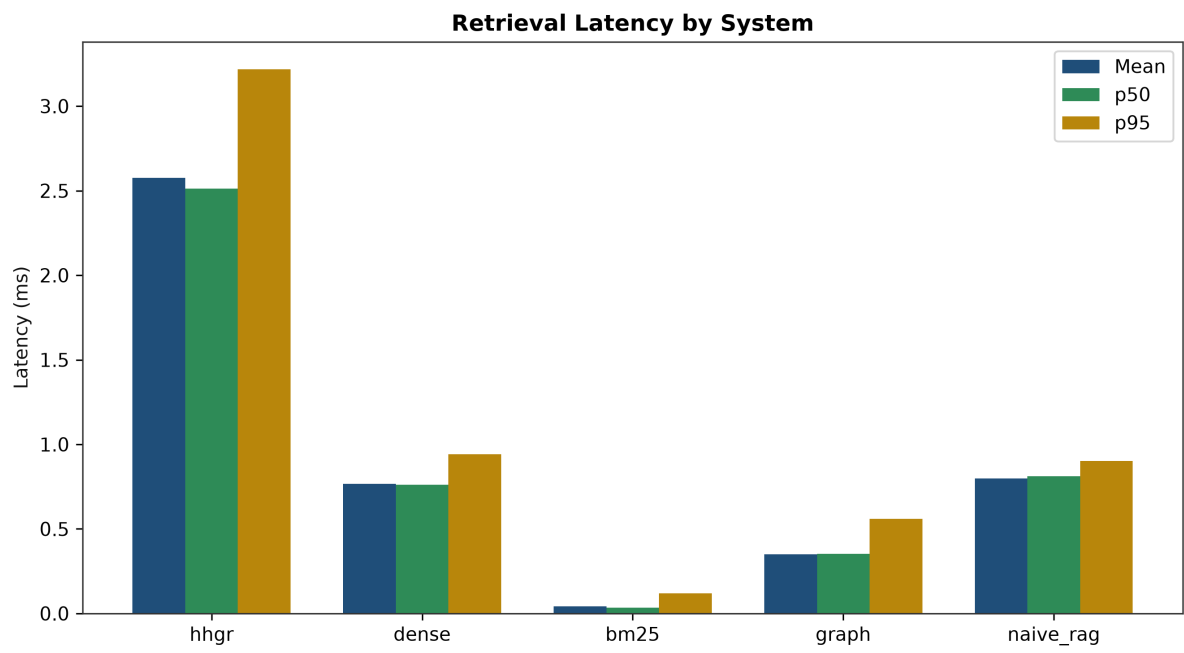


Hierarchy Tree — 0940d367554383c5.json



Knowledge Graph Example — Indian Contract Act, 1892 (PART_OF + citation edges)





RAGAS-style Generation Metrics (offline surrogate)

